

Switching Circuits Laboratory (CS29002)

Assignment 5 (11-02-2026)

- a) Using a TTL 4-bit adder IC (74LS83) and other required gates, design a 4-bit adder/subtractor circuit. Apply the inputs via on-board switches and observe the outputs via on-board LEDs. Verify the operation with the following data inputs (assume that the input data are encoded as 4-bit 2's complement numbers):
- i) $3 + 4$
 - ii) $6 - 7$
 - iii) $(-3) + (-5)$
 - iv) $(-8) - (-2)$

Instructions:

- 1) *Draw the circuit diagram on paper, following the suggested conventions. Following this, you can issue the components and start with the realization on the breadboard.*
- 2) *Prepare a laboratory report in PDF format and upload it to Moodle by the specified deadline. A single report has to be uploaded per group.*